Anish Neema

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Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Computer Engineering- GPA: 3.77/4.0 *Expected May 2028*
Relevant Coursework: Objects and Design, Algorithms, C++ and Java Programming, RISCV

Experience

Nuveen Technologies (TIAA) Jun. 2026 – Aug. 2026
Software Engineering Intern *Charlotte, NC*

- Engineered a trade verification pipeline using AWS Lambda, EventBridge, and AgentCore, enabling LLM agents to cross-reference RAG vector data with PostgreSQL records and saving 20 hours/week of manual review.
- Architected a serverless automation pipeline integrating Microsoft Graph REST API and pandas to validate 100+ email records daily and route stakeholder alerts, with deployments automated through Jenkins CI/CD.
- Containerized Python services with Docker, orchestrated deployments on Kubernetes, and provisioned cloud infrastructure using Terraform to support production releases during agile based sprint cycles.

Trustworthy Robotics Lab @ Georgia Tech Aug. 2026 – Current
Software Engineering Intern *Atlanta, GA*

- Implemented a C++ Model Predictive Control (MPC) solver for real-time trajectory planning under model uncertainty and disturbances, solving a 30-step, 12-state constrained problem in under 8 ms per step.
- Profiling the C++ implementation to locate computational bottlenecks, then porting them to CUDA kernels for GPU parallelism, achieving a 4x speedup over the CPU baseline and enabling 100 Hz control..

Algoverse Aug. 2024 – Mar. 2025
Machine Learning Engineering Intern *Remote*

- Created a Python LLM framework using Retrieval-Augmented Generation (RAG) and knowledge tracing to let models self-assess tool use before calling external APIs, cutting API calls by up to 90% on QA benchmarks.
- Built a synthetic data pipeline with JSON schemas and GPT-4 as a teacher model generating diagnostic pre-tests for cost-aware API decisions without fine-tuning, earning a **NAACL Insights Workshop** poster presentation.

Projects

Embedded Systems Game Engine | *C, ESP32, PlatformIO, Data Structures* Apr. 2026 – May 2026

- Built a real-time jetpack-style game in C on an ESP32 with PlatformIO, implementing a custom hash table with linked-list separate chaining to store and retrieve game-object state in O(1).
- Managed heap allocation and deallocation manually to prevent leaks across extended sessions, implemented collision-driven game-over state transitions, and profiled the loop to hold a consistent frame rate.

Gymseekr Mobile App | *React Native, AWS Amplify, Google Cloud, Full Stack* Jun. 2024 – Jun. 2025

- Engineered and launched a cross-platform fitness app using React Native, AWS Amplify, and Google Cloud, implementing CI/CD pipelines, location-based gym discovery, and scaling to 200+ users.
- Designed backend authentication and database infrastructure using AWS Amplify, managing user sessions, secure data access, and concurrent requests to support reliable operation during peak usage.

Leadership & Organizations

Vice President, Claude Builder Club @ Georgia Tech | *AI education* Apr. 2026 – Current

- Lead weekly workshops and project sprints for 50+ members, guiding teams in building AI-powered applications with the Claude API and Anthropic's ecosystem, focused on agentic system design.

C and Assembly Programming Teaching Assistant @ Georgia Tech | *C/C++* Aug. 2026 – Current

- Lead recitations and office hours for 100+ students on C and assembly, covering manual memory management with hash tables and linked lists, ISA datapath design, recursion, and pointer debugging.

Technical Skills

Languages & Frameworks: Java, Python, C/C++, CUDA, C#, JavaScript, SQL, RISC-V Assembly, React
Cloud & DevOps: AWS (Lambda, S3, EventBridge, AgentCore, Amplify), Docker, Kubernetes, Terraform
Data, ML & Tools: PostgreSQL, NoSQL, pandas, RAG & vector databases, Claude API, Jenkins CI/CD, Git